

Structured Training Program: Wi-Fi

Module 5: Advanced Features and Standard Extensions

1. What are the key features of Wi-Fi 6, 6E and 7 and how do they differ from previous standards like Wi-Fi 5 (802.11ac) ?

Key features of Wi-Fi 6, 6E, and 7 vs Wi-Fi 5 (802.11ac)

- a) Wi-Fi 5 (802.11ac): Operates only on 5 GHz, max 8 spatial streams, uses OFDM, MU-MIMO only on downlink, max 3.5 Gbps.
- b) Wi-Fi 6 (802.11ax):
 - OFDMA for multi-user efficiency
 - Uplink + Downlink MU-MIMO (8x8)
 - Target Wake Time (TWT) for IoT
 - BSS Coloring to reduce interference
 - 9.6 Gbps theoretical max
 - Operates on 2.4 GHz and 5 GHz
- c) Wi-Fi 6E: All Wi-Fi 6 features + extends into the 6 GHz band, adding up to 1200 MHz of new spectrum and 14 additional 80 MHz channels or 7 × 160 MHz channels.
- d) Wi-Fi 7 (802.11be):
 - Multi-Link Operation (MLO)
 - 320 MHz channel width
 - 4096-QAM modulation
 - Up to 16 spatial streams
 - 46 Gbps theoretical max
 - Operates across 2.4, 5, and 6 GHz simultaneously

2. Explain the role of OFDMA in Wi-Fi 6 and how it improves network efficiency.

Role of OFDMA in Wi-Fi 6

OFDMA (Orthogonal Frequency Division Multiple Access) divides a channel into smaller Resource Units (RUs), allowing the AP to serve multiple clients simultaneously in a single transmission window. Unlike Wi-Fi 5's OFDM which served one device at a time per channel, OFDMA enables parallel uplink and downlink transmissions, reducing latency, improving efficiency in dense environments, and minimizing idle airtime.

3. Discuss the benefits of TWT in Wi-Fi 6 for IoT devices.

Benefits of Target Wake Time (TWT) for IoT

- TWT allows an AP to schedule specific times for IoT devices to wake up and communicate, then return to sleep. Benefits include:

- Battery life extension — devices sleep longer between transmissions
- Reduced contention — fewer devices competing simultaneously
- Predictable communication — scheduled intervals improve reliability
- Scalability — supports thousands of low-power sensors in one network

4. Explain the benefits of 6 GHz frequency band in Wi-Fi 6.

Significance of the 6 GHz Band in Wi-Fi 6E

The 6 GHz band (5.925–7.125 GHz) adds up to 1200 MHz of clean, uncluttered spectrum. Key significance:

- No legacy devices allowed — less interference
- Supports wider channels (160 MHz/320 MHz) for high throughput
- More non-overlapping channels available
- Ideal for high-bandwidth, low-latency applications (AR/VR, 4K/8K streaming)
- Reduces congestion on the already crowded 2.4 GHz and 5 GHz bands

5. Compare and contrast Wi-Fi 6 and Wi-Fi 6E in terms of range, bandwidth, and interference.

Feature	Wi-Fi 6	Wi-Fi 6E
Frequency	2.4 & 5 GHz	2.4, 5 & 6 GHz
Range	Better (lower freq penetrates walls)	Shorter (6 GHz attenuates faster)
Bandwidth	Up to 160 MHz channels	Up to 160 MHz (more available channels)
Interference	Comparatively Higher (shared with legacy devices)	Much lower (6 GHz is clean spectrum)
Channel availability	Limited non-overlapping channels	Many more non-overlapping channels
Best use case	General coverage, mixed environments	High-density, high-throughput, short range

6. What are the major innovations introduced in Wi-Fi 7 ?

- Multi-Link Operation (MLO): Simultaneous use of multiple bands/channels
- 320 MHz channel width: Double Wi-Fi 6E's maximum
- 4096-QAM: Higher modulation density = 20% more throughput vs 1024-QAM
- Multi-RU (MRU): Assign multiple resource units to a single device
- 16 spatial streams: Up from 8 in Wi-Fi 6/6E
- Enhanced MU-MIMO: Supports more simultaneous users
- 46 Gbps theoretical peak throughput

7. Explain the concept of MLO and its impact on throughput and latency.

MLO allows a device to simultaneously connect and transmit/receive across multiple bands (e.g., 5 GHz + 6 GHz) using a single logical connection. Impact:

- Throughput: Aggregates bandwidth across links, significantly increasing effective data rates
- Latency: Traffic can be dynamically shifted to the least congested link, reducing delays
- Reliability: If one link degrades, traffic seamlessly moves to another (link redundancy)
- Load balancing: The AP intelligently splits traffic across links based on conditions

8. What is the purpose of 802.11 k? How do they aid in roaming?

a) 802.11k (Radio Resource Management):

Enables clients to request a neighbor report — a list of nearby APs with signal data. Instead of scanning all channels, the client knows exactly where to roam next, speeding up decisions.

b) 802.11v (BSS Transition Management):

Allows the AP to actively steer clients to a better AP by sending a BSS Transition Management Request. The AP can suggest or enforce roaming when a client's signal degrades or load balancing is needed.

Together, they reduce roaming time from hundreds of milliseconds to tens of milliseconds.

9. Explain the concept of Fast BSS Transition (802.11r) and its benefits in mobile environments.

802.11r pre-authenticates a client with the target AP before it roams, caching the security keys (PMK-R1) at neighboring APs via the DS (Distribution System). When the client roams, it skips the full 802.1X/EAP authentication handshake.

Benefits:

- Reduces roaming handoff time from ~500ms to under 50ms
- Critical for voice (VoIP) and video calls that drop if handoff exceeds ~150ms
- Seamless experience in hospitals, warehouses, and campuses where users are constantly moving

10. How do 802.11 k/v/r work together to provide seamless roaming in enterprise networks?

These three protocols form a complete roaming framework:

ProtocolRole802.11kWhere to go — provides neighbor AP list802.11vWhen to go — AP pushes the client to roam proactively802.11rHow to go fast — pre-authenticates, enabling sub-50ms handoff

Protocol	Role
802.11k	Where to go — provides neighbor AP list
802.11v	When to go — AP pushes the client to roam proactively
802.11r	How to go fast — pre-authenticates, enabling sub-50ms handoff

Workflow example:

- a) AP detects client signal weakening, uses 802.11v to recommend a better AP
- b) Client uses 802.11k neighbor report to identify the target AP without full channel scan
- c) Client executes fast roam using 802.11r, completing the handoff in milliseconds

This combination is essential in enterprise networks supporting VoIP phones, barcode scanners, medical devices, and mobile workers requiring uninterrupted connectivity.